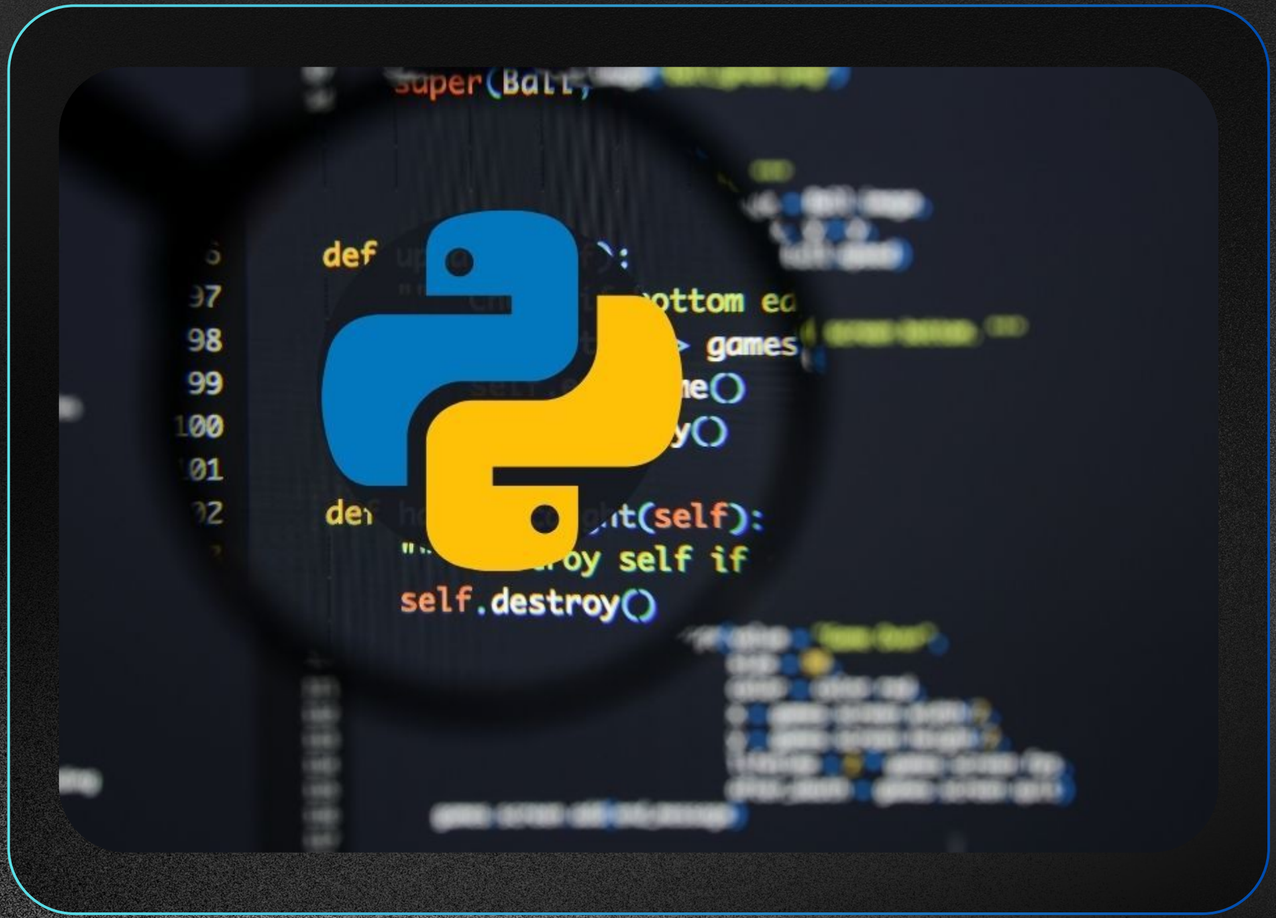
BINARY SEARCH AND PROGRAM IN PYTHON

EDS Assignment 1



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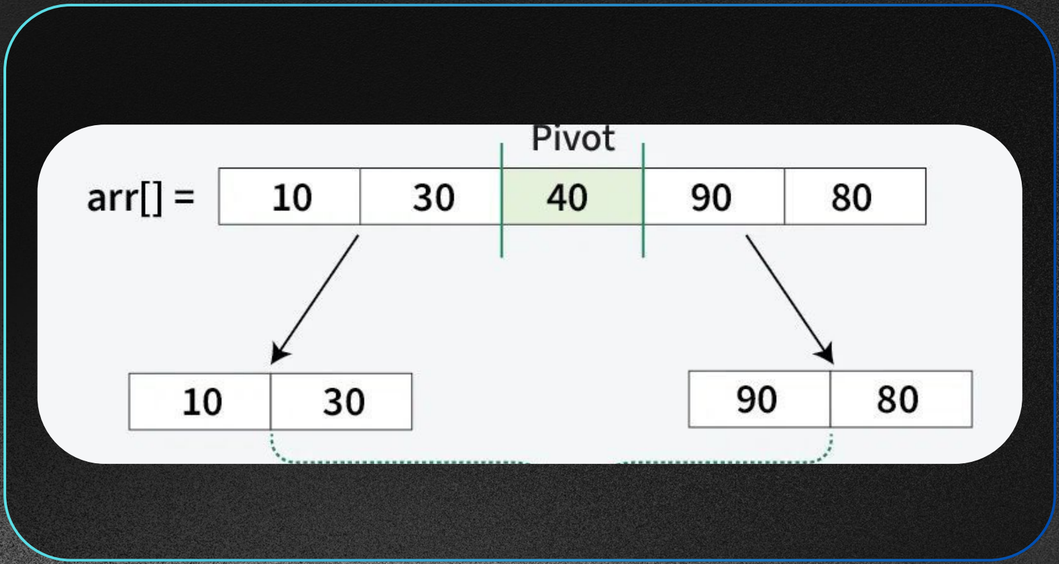
INTRODUCTION

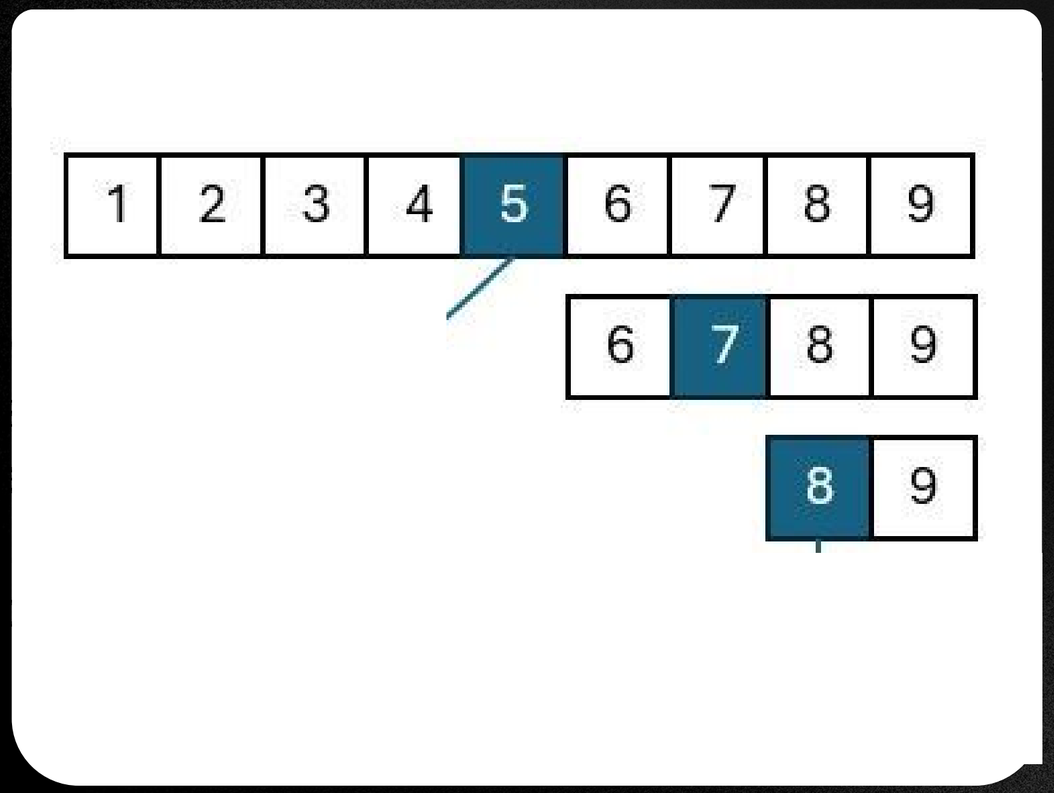
- Searching means finding an element in a list
- Used in real-life applications like Google search, contacts, etc.
- There are many searching techniques
- Binary Search is one of the fastest methods

WHAT IS BINARY SEARCH?

- Binary Search is an efficient searching algorithm
- Works only on sorted lists
- It divides the list into halves
- Compares the middle element



HOW BINARY SEARCH WORKS

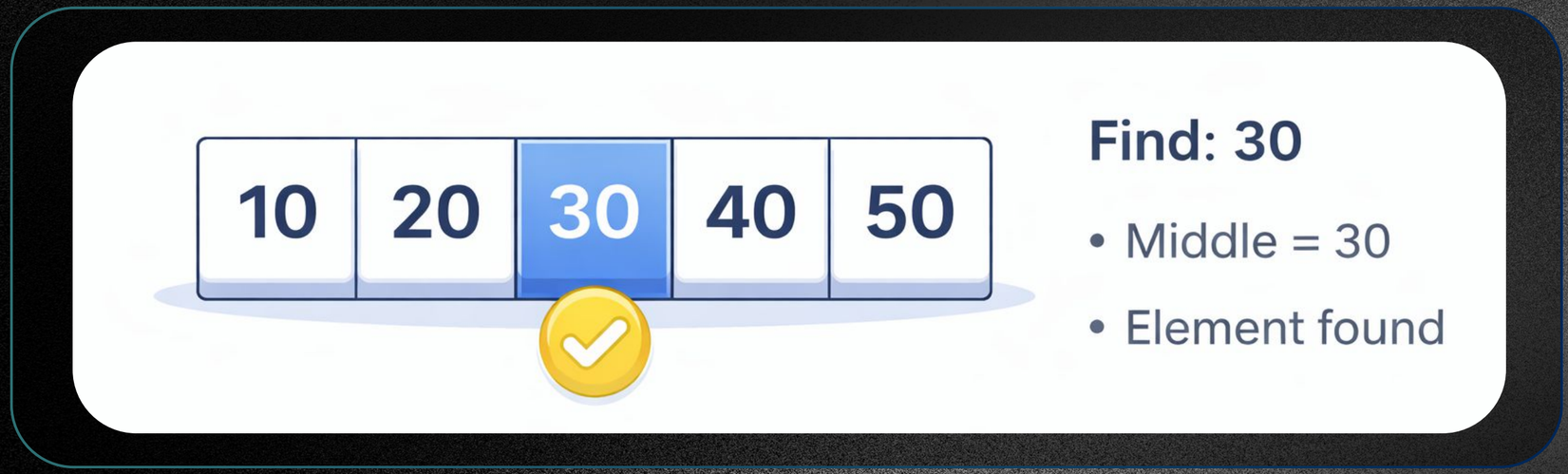


1. Find middle element
2. If target = middle → Found
3. If target < middle → search left side
4. If target > middle → search right side
5. Repeat until found or list ends

EXAMPLE

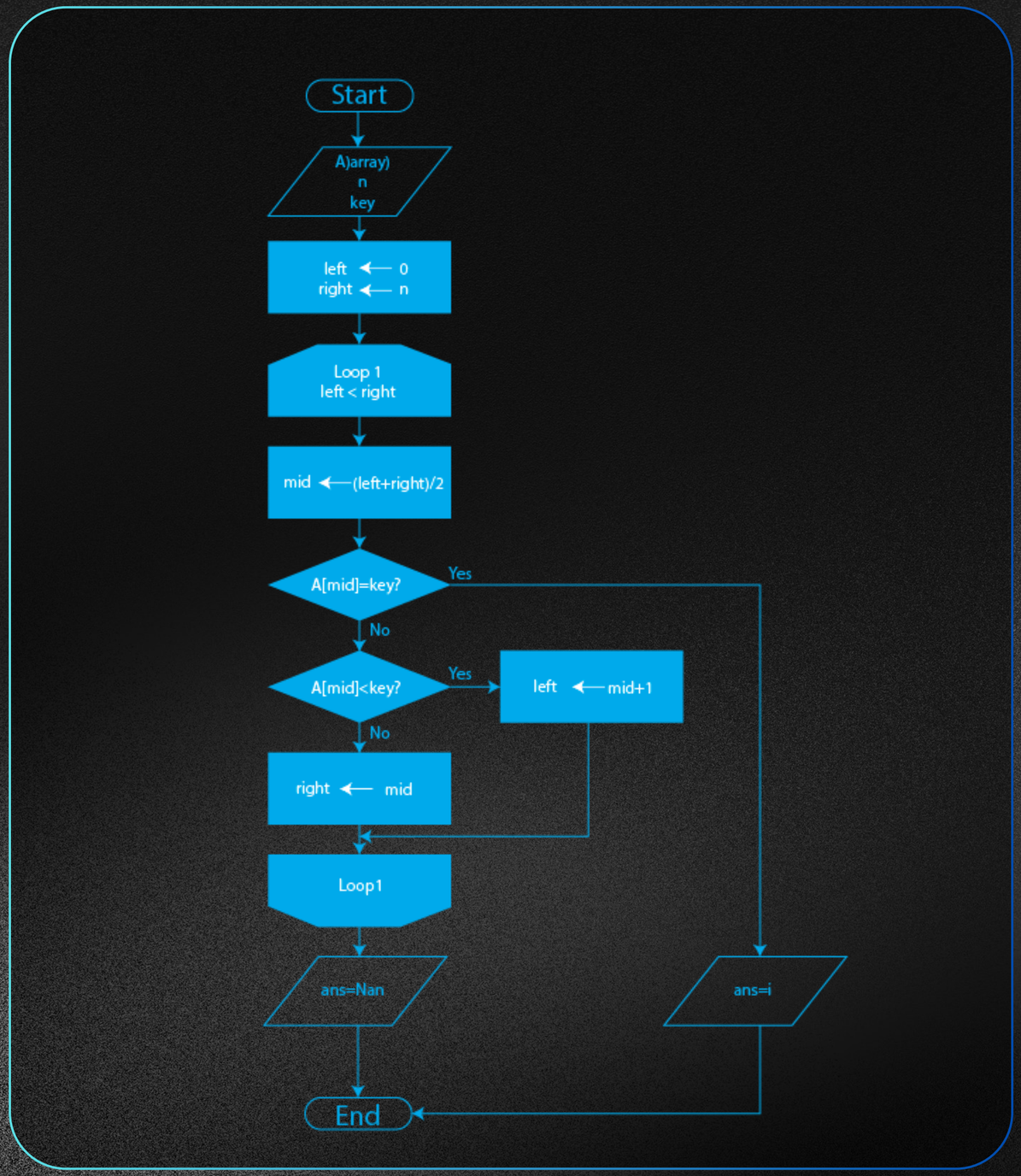
Example :-
List: [10, 20, 30, 40, 50]
Find: 30

Steps:
- Middle = 30
- Element found



ALGORITHM

Start
Set low = 0, high = n-1
While low <= high:
  mid = (low + high) // 2
  If element == mid → found
  If element < mid → high = mid - 1
  If element > mid → low = mid + 1
End



PYTHON PROGRAM

Python Online Compiler

main.py | Output

```python
def binary_search(arr, target):
    low = 0
    high = len(arr) - 1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == target:
            return mid

        elif arr[mid] < target:
            low = mid + 1

        else:
            high = mid - 1

    return -1

```

```python
        else:
            high = mid - 1

    return -1

arr = [10, 20, 30, 40, 50]
target = 30

result = binary_search(arr, target)

if result != -1:
    print("Element found at index", result)
else:
    print("Element not found")
```

OUTPUT

Programiz
Python Online Compiler

Programiz PRO

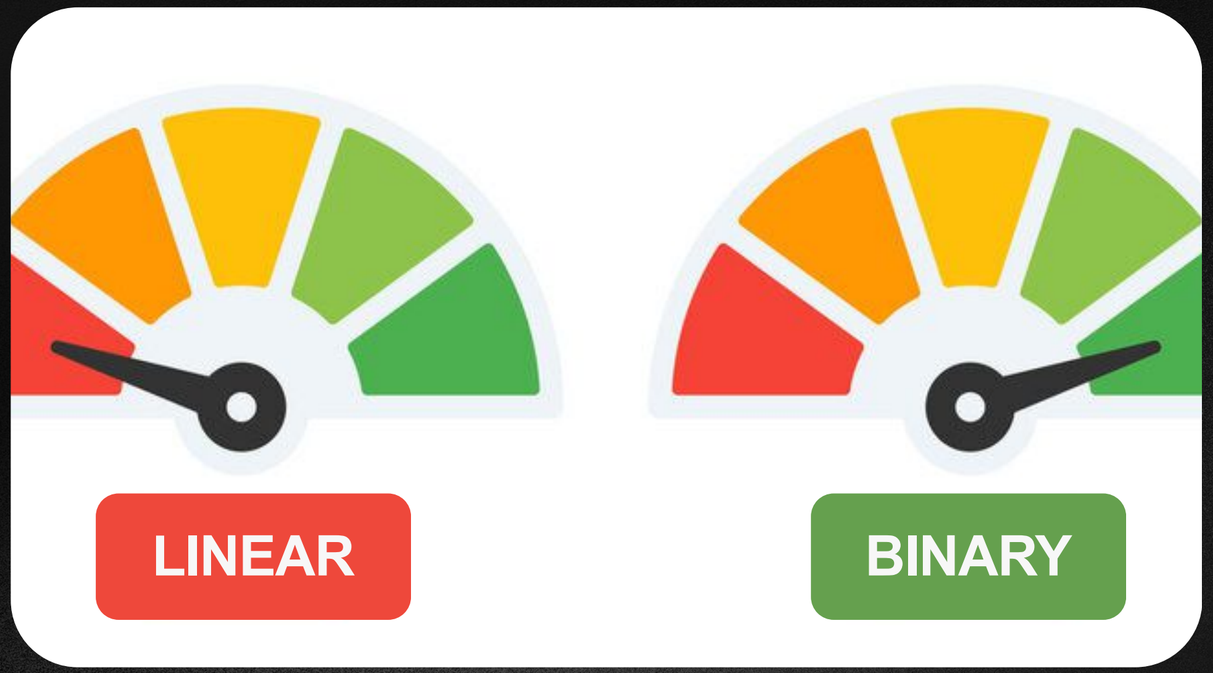
main.py | Output

```
Element found at index 2

=== Code Execution Successful ===
```

ADVANTAGES

- Very fast compared to linear search
- Efficient for large datasets
- Reduces search time



DISADVANTAGES

- Works only on sorted data
- Slightly complex compared to linear search

APPLICATIONS

- Searching in databases
- Library systems
- Search engines
- Finding values in large datasets

CONCLUSION

- Binary Search is fast and efficient
- Requires sorted data
- Widely used in programming

THANK YOU !!